PROJECT AND TEAM INFORMATION

Project Title

EcoLife – Ecosystem Impact Analysis and What-If Decision System

Student / Team Information

Team Name:	**ECOLIFE**
Team ID :	**DSCPP-III-2026-T169**
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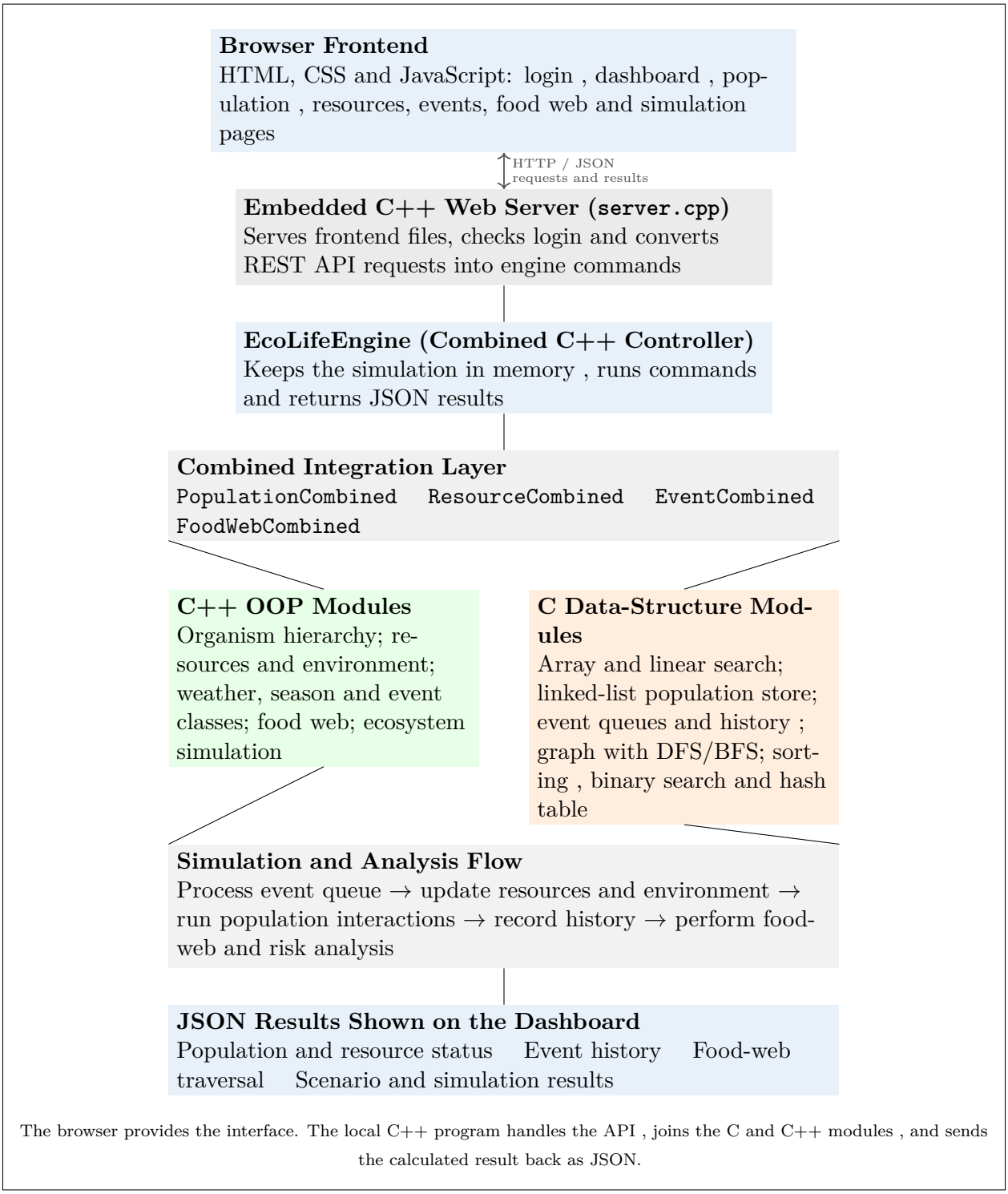
PHASE-II PROJECT PROGRESS DESCRIPTION(35 pts)

Project Abstract

EcoLife is an educational , rule-based ecosystem simulation and impact-analysis project. It is intended to help a user observe how one disturbance can move through connected species over virtual days. For example , a drought may reduce plant growth , herbivores may then have less food , and carnivores may later be affected by the reduced herbivore population. The proposed ecosystem uses a small set of plants , herbivores , carnivores and birds so that these links can be followed clearly during a demonstration.
The simulation will use understandable rule for food availability , feeding , reproduction , predation , death , weather and selected events such as fire or predator removal . It's purpose is to show the direction of a possible effect , not to make a scientific prediction about a real ecosystem. Phase2 planning extends the original population idea by adding food-web impact tracing, event records, scenario comparison and a simple simulated risk ranking. The users will be able to compare a baseline situation with a chosen event and see which species are directly or not directly affected.
The Phase2 now contains an initial integrated prototype with a browser frontend, a small local server, Cpp simulation classes and C data-structure modules. The next work is to build and check the complete program on the presentation system , test each feature with fixed scenarios , and improve the explanation of the results shown to the user.

Updated System Architecture (High-Level Diagram)

(Phase-II architecture prepared from the current project source.)



The browser provides the interface. The local C++ program handles the API , joins the C and C++ modules , and sends the calculated result back as JSON.

Tasks Completed

Task Completed	Team Member
Phase-II scope refinement : converted the idea into a rulebased impact-analysis and what-if system.	Entire team
Prepared the proposed Cpp organism hierarchy and the responsibility of the ecosystem , environment and event classes.	Unnati Gupta and Amritaanshu Dheeman
Mapped the planned C data structures to the project purpose , including event queues , species lookup , risk ranking and the food web graph.	Rishabh Budakoti
Prepared the initial Phase 2 project structure : browser frontend , embedded local server , combined engine and separate C and C++ module folders.	Entire team
Prepared the system architecture , report structure , presentation material and initial demonstration plan .	Adarsh Mishra and Amritaanshu Dheeman

Challenges/Roadblocks

The main challenge is choosing rules that are simple enough for a student project but still show a believable chain of effects . A highly detailed ecological model would require real field data , many parameters and complex calculations . On the other hand , a very small model may hide the connection between food availability and population change . The team plans to use a fixed group of species and clearly document every rule, thresholld and event effect. This will make it possible to explain why a populations changed in a particular scenario.
A second challenge is the connection between the Cpp objects and C data structures. The current source keeps the C modules and Cpp classes in separate folders , while the combined layer brings them together. This separation is useful , but it means the team must carefully check how population values , event records and food-web data move between modules. The C++ classes should manage behaviour , and the C layer should manage the records needed for searching and analysis . The next step is to build the prototype on the presentation system and check a small baseline case before adding more event cases. Empty , normal and unusual inputs will be tested before the complete simulation is demonstrated.

Tasks Pending

Task Pending	Team Member(s)
Build the current Cpp classes, ecosystem controller and daily simulation cycle as one runnable program.	Unnati Gupta, Adarsh Mishra
Run and verify the C data structures modules for linked lists, queues , priority queue , hash table and graph.	Rishabh Budakoti
Check environmental events, foodweb traversal, risk ranking and population analysis using fixed scenarios.	Entire team
Prepare unit tests , scenario checks , final documentation and the working demonstration.	Amritaanshu Dheeman , Entire team

Project Outcome/Deliverables

The current Phase 2 deliverables include the refined project scope , module design , Cpp class hierarchy , data structures mapping , updated architecture, report, presentation and the initial project source. The source is organised into a browser frontend, a local Cpp server and engine , C modules , Cpp modules, combined integration files and test material. This structure gives the team a practical base for building and demonstrating the project.
The next technical deliverables are a verified runnable build , checked environmental events, population and event history , foodweb traversal , simulated risk ranking and scenario comparison . A scenario report will show the selected event and the observed changes over simulated days. The final submission will include the source code , build instructions , test-records and a working demonstration . These items still need final validation before they can be presented as results .

Progress Overview

The design work is complete, and an initial Phase 2nd code structure has also been prepared . The project is divided into a browser interface, a local server and engine , C data-structure modules , C++ OOP modules and combined integration files. This is useful progress because the intended flow of data is now visible in the project files rather than only in the written design .
The main remaining work is now just of verification . The team needs to build the complete program , run the module checks , and confirm that the frontend and engine give the expected result for the same given input. A baseline scenario will be checked firstly , followed by drought , fire and predator-removal cases. Test records , clear screenshots and a rehearsed demonstration are still needed. This order will help the group find a module-level issue before it affects the complete made simulation .

Codebase Information

The current Phase 2nd source is arranged in separate folders for the frontend , the C modules, the C++ modules , the combined integration layer , tests and documentation. The main program starts a local server, while the frontend uses Http and Json to request population , resource , event, food-web and simulation data . The combined folder contains the engine and the code that connects the C and C++ parts of the entire project .
The project still needs a shared repository link and final commit history before submission . The team will add a short README with build steps , member responsibilities and the final folder structure. The stable version will be kept on the main branch , while any correction to an individual module can be checked separately before it is merged . This will make the final changes and testing work easier to track and make any changes if required.

Testing and Validation Status

Test Type	Status	Notes
Design review	Completed	The team reviewed the module plan, C/C++ responsibility split and data-structure purpose.
C++ class compilation	Pending check	Class and engine source files are present; the complete build needs verification on the presentation system.
C data-structure tests	Pending check	C module demos and test material are present; empty, single-item and duplicate cases still need to be recorded.
C/C++ integration check	Pending check	The combined engine connects the two layers; the full program must be built and exercised end to end.
Scenario validation	Planned	Drought, fire and predator-removal cases will be compared with one baseline scenario.

Deliverables Progress

- Phase2 report, presentation, refinement and architecture: **completed**.
- C++ object model, C data structure modules and combined engine source: **prepared ; build verification is pending**.
- Browser frontend, local server and five module pages: **prepared ; end to end checking pending**.
- Automated tests, recorded scenario results and performance measurements: **pending**.
- Working demo, repository link and final project report: **still needed to be completed**.

References

1. B. Stroustrup, The C++ Programming Language, 4th ed, Addison-Wesley professional, 2013
2. B.W. Kernighan and D.M. Ritchie, The C Programming Language, 2nd ed., Prentice hall, 1988
3. T.H. Cormen, C. E. Leiserson, R. L. Rivest and C. Stein, Introduction to Algorithms, 4th ed., MIT press, 2022
4. OpenStax, Biology 2e, 46.1 Ecology of Ecosystems, Rice University
5. OpenStax, Biology 2e, 45.6 Community Ecology,Rice University